

A Low-Variability Vigour-Related fMRI Network in Parkinson's Disease Is Distinct From Task Activation and Sensitive to Dopaminergic State

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2 ABSTRACT

3 Trial-wise variability in movement vigour is common in Parkinson's disease (PD), but behaviour
 4 alone cannot show whether this instability arises in a central vigour-related neural signal or during
 5 later translation into movement. We analysed trial-wise fMRI and reaction time (RT) from 18
 6 participants with PD performing a reward-cued ballistic hand-squeeze task during OFF- and ON-
 7 medication sessions, with galvanic vestibular stimulation (GVS) delivered in randomized blocks.
 8 Single-trial neural responses were estimated with GLMs and entered into a constrained
 9 optimization model designed to recover a spatially coherent projection that was task engaged,
 10 RT linked, and stable across consecutive trials. The optimized projection was highly reproducible
 11 across cross-validation folds ($ICC = 0.978$) and showed lower consecutive-trial variability than
 12 RT in subject-level summaries derived from 70 retained runs (mean behaviour-minus-projection
 13 difference = 7.5, 95% CI [2.2, 12.8], $p = .008$). The final network contained 8,182 voxels and
 14 included motor, striatal, cerebellar, thalamic, temporal-limbic, orbitofrontal, parietal, and contextual
 15 regions. It overlapped weakly with the standard task-activation map (835 shared voxels; Dice
 16 = 0.056; Jaccard = 0.029), indicating that the recovered signal was not simply general task
 17 activation. Dopaminergic medication altered network organization, with a significant increase in
 18 local within-ROI coupling in the vigour network ($\Delta_{ON-OFF} = +0.0147$, $p = 0.006$) and no reliable
 19 between-ROI change ($\Delta_{ON-OFF} = -0.0006$, $p = 0.723$). Exploratory GVS analyses showed
 20 heterogeneous sham-referenced connectivity changes within the network, including 156 mutual-
 21 information and 194 Spearman edge changes. These findings show that a relatively stable central
 22 vigour-related signal can be recovered in PD, while dopaminergic state and vestibular stimulation
 23 modulate the network through which that signal is expressed.

24 **Keywords:** Parkinson's disease, fMRI, motor vigour, reaction-time variability, dopaminergic medication, galvanic vestibular stimulation,
 25 functional connectivity

1 INTRODUCTION

26 Movement vigour refers to how quickly and energetically an action is initiated and carried out. It is not a
 27 fixed property of the limb: the same person can perform the same action faster or slower across trials, and
 28 different people often express different levels of speed and force during the same task (Berret et al., 2018).
 29 Human fMRI work links reach speed to a distributed set of cortical and subcortical regions, including
 30 motor and premotor cortices, supplementary motor area, basal ganglia, thalamus, parietal cortex, and
 31 cerebellum (Shirinbayan et al., 2019). ECoG and basal-ganglia studies further support movement-related
 32 cortical activity and subcortical contributions to force amplitude (Anderson et al., 2012; Spraker et al.,
 33 2007). These findings suggest that vigour is not merely a behavioural output, but a regulated feature of
 34 action controlled by interacting neural circuits.

Dopamine is central to this regulation. Basal-ganglia output can change movement speed and amplitude without necessarily changing which action is selected (Turner and Desmurget, 2010), and striatal pathway manipulations can speed up or slow down movement in a dopamine-dependent manner (Yttri and Dudman, 2016). Reward and motivational value also influence vigour: actions associated with higher value are often initiated faster and executed with greater speed (Kawagoe et al., 1998; Reppert et al., 2015; Summerside et al., 2018). Parkinson's disease (PD), which is characterized by dopaminergic loss in basal-ganglia circuits, is therefore a useful model for asking how dopaminergic state shapes the consistency of vigour expression.

A core motor symptom of PD is bradykinesia, usually described as slower and smaller movement. The problem, however, is not simply that patients cannot move quickly. When speed is strongly encouraged, patients with PD can sometimes produce rapid movements (Mazzoni et al., 2007). This suggests that PD may disrupt how vigour is regulated and deployed from one trial to the next. Behavioural studies report larger short-term response-time fluctuations in PD (MacDonald et al., 2025), and earlier work found increased endpoint variability that could be partly reduced when participants slowed down or relied on visual guidance (Sheridan and Flowers, 1990). Motor output is also less stable: motor-unit discharge can become more variable in mild-to-moderate PD even when mean discharge rates are relatively preserved (Wilson et al., 2020). Dopamine loss may therefore affect both average movement features and the consistency with which those features are expressed (Lee et al., 2015).

The key difficulty is that behavioural variability has several possible sources. Some movement variability is useful, helping the motor system explore actions, adapt after errors, and learn from reward (Pekny et al., 2015; Wu et al., 2014). Some variability reflects the neural state before movement; preparatory activity in motor and premotor cortex can predict the speed of the upcoming reach (Churchland et al., 2006). Another source is motor noise (Faisal et al., 2008), especially signal-dependent noise, where larger motor commands generate greater variability and make fast or forceful movements harder to control accurately (Harris and Wolpert, 1998). In PD, such downstream noise may be behaviourally costly: signal-dependent noise has been linked to effort costs in the more affected arm, and reducing this noise increased patients' willingness to use that arm (Salimpour et al., 2015). High motor noise can also weaken the link between an action and its outcome, impairing learning (Therrien et al., 2016).

Because these sources occur together, reaction time alone cannot reveal whether variable vigour reflects instability of a central neural signal or noise added later in the motor pathway. This distinction matters for PD. If the central vigour signal itself is unstable, then bradykinesia and inconsistency may reflect impaired neural coding. If a relatively stable central signal still tracks trial-wise behaviour while behavioural output fluctuates, then part of the instability may arise downstream of that signal or during its state-dependent translation into movement. Trial-wise fMRI provides a way to test this idea because it can estimate neural responses for individual trials and relate those responses to moment-to-moment behaviour rather than only to average task activation (Dinstein et al., 2015; Garrett et al., 2013; Haar et al., 2017).

The present study was designed for this central-versus-downstream question while also testing whether the recovered network is sensitive to dopaminergic and vestibular modulation. We analysed a reward-cued ballistic hand-squeeze task in PD participants scanned OFF and ON dopaminergic medication. During the task, participants also received block-wise galvanic vestibular stimulation (GVS), an exploratory perturbation relevant to motor-control and movement-disorder stimulation research. We developed a constrained optimization framework to recover a neural projection that was task engaged, linked to trial-wise RT, spatially coherent, and low in consecutive-trial variability. We then asked four questions: (i) whether the projection was reproducible and less variable than behaviour, (ii) whether its anatomical distribution was distinct from standard task activation, (iii) whether dopaminergic medication changed its network organization, and (iv) whether GVS perturbed connectivity within the recovered network.

2 MATERIALS AND METHODS

2.1 Participants and medication-state design

Eighteen participants with Parkinson's disease were included in the final analysis (4 female, 14 male; mean age 67.22 ± 7.03 years, range 55–82). Fourteen participants were right-handed, and four were left-handed. Participants performed the task with their dominant hand, except for one right-handed participant who used the left hand because tremor in the dominant hand was severe. Data from Subject 17, session

1 (OFF medication), were excluded because one recording run was excessively noisy. Exclusion criteria included prior neurosurgical treatment for PD, other neurological or major medical conditions, dementia severe enough to preclude informed consent, severe levodopa-induced dyskinesia incompatible with MRI, and standard contraindications to MRI. Clinical severity was assessed using the MDS-UPDRS Part III (Goetz et al., 2008); detailed medication-state and domain summaries are reported in Supplementary Tables S1 and S2. All participants provided written informed consent, and the study was approved by the Clinical Research Ethics Board at the University of British Columbia.

Participants completed two medication sessions on the same day: OFF medication and ON medication. They were instructed to stop dopaminergic medication at least 12 h before recording. The session order was fixed across participants: the OFF-medication session was always completed first, after which participants took their dopaminergic medication and completed the ON-medication session approximately 1 h later. Because OFF sessions always preceded ON sessions, medication effects could be partly confounded with practice, fatigue, time-on-task, or scanner drift. We therefore treated medication results as state-associated effects rather than as effects from a fully randomized medication manipulation.

2.2 Behavioural task and GVS protocol

During fMRI, participants performed a ballistic hand-squeeze task implemented in MATLAB with Psychtoolbox (Brainard, 1997) while lying supine in the scanner and viewing the task through a mirror display (Figure 1A). On each trial, a visual cue indicating reward value was presented for 2 s, followed by a variable delay of 2–3.5 s. A moving balloon then appeared as the Go signal. Participants were instructed to squeeze a mechanical bulb as quickly and forcefully as possible when the Go signal appeared, and successful responses were followed by a visual burst of the balloon. Reaction time (RT), defined as the interval between Go-signal onset and squeeze initiation, was used as the primary trial-wise index of motor vigour because it was the most robust behavioural measure in this dataset.

Each medication session consisted of two task runs, and each run included nine blocks of 10 trials. Most trials were rewarded, with low- and high-reward conditions counterbalanced across runs, and 10% were catch trials in which participants were required to withhold their response to avoid a penalty. Task difficulty was adjusted online by varying the response window based on recent hit rate, with the aim of maintaining accuracy at or above 65%. A 20 s rest period was provided after every three blocks. Behavioural quality-control and reward-related RT summaries are reported in Supplementary Figures S1–S3.

GVS was delivered continuously throughout each run using bilateral mastoid electrodes. Each run contained one sham block and eight active amplitude-modulated GVS blocks. The active waveforms crossed carrier frequency (75 or 125 Hz) with envelope frequency (0, 10, 16, or 20 Hz), producing GVS1–GVS8. The order of the nine stimulation conditions was randomized independently for each run. The GVS analyses were considered exploratory because stimulation was presented in short blocks, observations from the same participant were repeated, and no dedicated washout interval separated every adjacent block. Full stimulation parameters, electrode placement, thresholding, and block-order details are provided in Supplementary Methods.

2.3 MRI acquisition, preprocessing, and trial-wise beta estimation

Structural and functional MRI data were acquired on a Philips 3T Ingenia Elition X whole-body MRI scanner upgraded to an MR7700 system (Philips Healthcare, Best, The Netherlands) using a 32-channel head coil. For each participant, a T1-weighted structural image was acquired, followed by two task-based fMRI scans per medication session. Functional MRI was acquired using EPI. The first three participants were scanned with a multi-echo EPI protocol, whereas the remaining participants were scanned with a single-echo EPI protocol. This protocol difference is reported explicitly because it may have affected signal-to-noise, temporal sampling, or trial-wise beta estimation. Full sequence parameters are provided in Supplementary Methods.

Image preprocessing was performed in FSL (Smith et al., 2004). Structural images underwent bias-field correction and brain extraction. Functional images were corrected for head motion and susceptibility-induced distortion, high-pass filtered, brain extracted, denoised using ICA-AROMA (Pruim et al., 2015), spatially smoothed with a 5 mm full-width at half-maximum Gaussian kernel, registered to each participant's structural image, and normalized to MNI152 space. The scanner automatically discarded four

dummy volumes before saved image acquisition, so no additional initial volumes were removed during preprocessing.

Task-related responses for individual trials were estimated using GLMsingle (Kay et al., 2013; Prince et al., 2022). Analyses were performed separately for each subject-session, with the two task runs analysed jointly while preserving run identity in the design matrix. Go-signal onsets were modeled using 2-s event regressors; catch trials were excluded because no squeeze response was required. The mean CSF time series was included as a nuisance regressor. GLMsingle was run with HRF fitting, GLMdenoise, and fractional ridge regression enabled. After robust beta-value cleaning and group-level coverage filtering, the final group beta matrix contained 91,769 voxels and 5,463 retained Go trials. Detailed beta-cleaning thresholds, outlier rules, and coverage calculations are reported in Supplementary Methods.

2.4 Standard task-activation reference map

To provide a conventional task-engagement reference, we ran a standard FSL FEAT GLM separate from the trial-wise GLMsingle model. All valid Go trials were modeled with one event regressor defined at the Go-signal onset, with duration 2 s and amplitude 1. First-level maps tested the positive effect of task activity, runs were combined within subject-session using fixed effects, and subject-session maps were entered into a group mixed-effects model using FLAME stage 1 (Woolrich et al., 2004; Beckmann et al., 2003). Group maps were thresholded using clusters formed at $Z \geq 3.1$ with corrected cluster significance $p \leq 0.05$ (Worsley, 2001). This thresholded map was used only as a standard task-activation reference, not as the main model. The full standard GLM map and comparisons with GLMsingle activation maps are shown in Supplementary Figures S4 and S5.

2.5 Constrained optimization framework

We used a group-level optimization procedure to identify a vigour-related brain network (Figure 1B,C). The purpose was to find a voxel-weight vector that combined whole-brain activity into one scalar neural projection per trial. For the trial-wise beta matrix X_β , the projected signal was

$$y = w^\top X_\beta, \quad (1)$$

where w is the weight vector and y is the projected trial-wise neural signal. The same projection framework was also applied to trial-window BOLD features to constrain local trial-to-trial instability. Because direct optimization over all voxels was computationally impractical, beta and BOLD data were reduced with expectation-maximization principal component analysis (EMPCA) (Roweis, 1997). An 80% cumulative explained-variance threshold retained 14 components for optimization; final weights were projected back to voxel space after model fitting.

The model encoded four assumptions. First, the selected signal should come from voxels engaged by the task. Second, the projected beta signal should be behaviourally relevant to RT. Third, neighbouring voxels should have similar weights, producing a spatially coherent map rather than isolated scattered voxels. Fourth, the projected signal should show low consecutive-trial variability. Consecutive-trial variability was used rather than total variance because it preserves trial order and directly penalizes abrupt local jumps while allowing slower task-related changes across runs or blocks. Run and block boundaries were excluded from this penalty because they do not represent continuous local temporal dynamics.

The optimization combined the task-engagement, BOLD variability, beta variability, and spatial-smoothness penalties into

$$C_{\text{total}} = \alpha_{\text{task}} C_{\text{task}} + \alpha_{\text{BOLD}} C_{\text{BOLD}} + \alpha_\beta C_\beta + \alpha_{\text{smooth}} C_{\text{smooth}}. \quad (2)$$

Behavioural relevance was represented by a correlation-based term linking the projected beta signal to RT. The final objective minimized penalties while maximizing behavioural relevance:

$$\min_w \frac{\gamma w^\top C_{\text{total}} w - w^\top C_b w}{w^\top C_d w} \quad \text{s.t. } \|w\|_2 = 1. \quad (3)$$

Here, C_b captures alignment between the projected beta signal and RT, C_d normalizes by the projected beta variance, the α values control the four penalty weights, and γ controls the overall balance between regularization and behavioural relevance. Full definitions of all terms and derivations are provided in Supplementary Methods.

The constrained optimization was solved using Sequential Least Squares Programming (SLSQP) (Nocedal and Wright, 2006; Kraft, 1988). Hyperparameters were selected with a coarse-to-fine cross-validation procedure. In the coarse stage, candidate penalty-weight combinations were evaluated over contiguous folds using held-out absolute correlation between the optimized neural projection and behaviour. After the best coarse setting was identified, a fine search tested multiplicative perturbations around that setting. The final setting was chosen based on held-out correlation, stability across folds, and small train-test differences. For each fold, the model was fit on training data, weights were projected back into voxel space, and final voxel maps and projected signals were obtained by averaging across folds.

2.6 Network definition and validation

The final vigour network was defined as voxels with weights above the 90th percentile. This threshold was chosen because it provided a stable compromise between spatial specificity and anatomical coverage: lower thresholds retained substantially more voxels, whereas stricter thresholds began to remove several anatomical groups. Threshold robustness is shown in Supplementary Figure S6.

We validated the solution in three ways. First, we assessed cross-validation reproducibility of the optimized voxel-weight maps using intraclass correlation (ICC). Second, we compared consecutive-trial variability of the projected neural signal with consecutive-trial variability of RT. The normalized consecutive-trial variability metric was

$$\sum_i \frac{(x_i - x_{i+1})^2}{x_i^2 + x_{i+1}^2}, \quad (4)$$

computed after excluding run and block boundaries. Runs contributed to the variability estimates, but inferential statistics used subject-level summaries rather than treating runs as independent observations. Third, we tested whether selected voxels were less variable than a motor-area control pool. The control pool was defined from bilateral motor-area atlas labels after removing selected voxels, and 1,000 size-matched samples were drawn without replacement.

2.7 Medication-state network analysis

After defining the vigour network, we tested whether dopaminergic state altered its organization. Medication state was not used during optimization, so this analysis was an independent characterization of the final network. Only participants with data in both medication states were included in paired analyses ($n = 17$). Selected voxels were assigned to atlas ROIs, and the original optimized voxel weights were retained when calculating ROI summaries so that voxels with larger model-derived weights contributed more strongly to the regional beta series. The same analysis was repeated using the standard task-activation map as a reference network.

Functional connectivity was defined from trial-wise ROI beta series; therefore, connectivity here refers to task-based co-fluctuation across trials, not resting-state connectivity. Pearson correlation measured linear co-fluctuation, and mutual information measured more general statistical dependence. Both measures are symmetric and undirected, so medication-related changes were interpreted as changes in statistical coupling rather than causal influence. We quantified whether network topology differed more between medication states than between repeated runs within the same state using correlation distance between vectorized ROI-edge patterns. This topology-control analysis is reported in Supplementary Figure S9.

The main medication analysis tested whether medication preferentially affected local within-ROI coupling or distributed between-ROI coupling. Between-region connectivity was the average Fisher- z -transformed Pearson correlation between all pairs of ROI beta series. Within-region connectivity was the average Fisher- z -transformed Pearson correlation between voxel beta series inside each ROI; ROI-level within-region values were averaged equally across ROIs so that larger ROIs did not dominate. The primary contrast

tested whether the ON–OFF change in within-ROI connectivity was larger than the ON–OFF change in between-ROI connectivity.

2.8 GVS-related network analysis

We next examined whether GVS modulated functional connectivity within the same ROI sets. For each subject, medication state, run, and GVS condition, trial-wise ROI beta estimates were grouped by stimulation block. The sham block served as the within-run baseline. Each active GVS condition was compared with the matched sham block from the same run.

Two complementary edge metrics were used. Quantile mutual information measured general statistical dependence after discretizing each ROI beta series into three rank-based quantile bins. Spearman's rank correlation measured monotonic rank-based association and provided a simpler, outlier-resistant comparison. For each active block, active-minus-sham edge changes were computed. For visualization and exploratory inference, block-level edge-change values were pooled across subjects, medication states, runs, and active GVS conditions. Each edge was tested against zero using a two-sided sign-flip permutation test with 5,000 permutations, followed by false-discovery-rate correction within metric and analysis scope. Because this analysis included repeated block-level observations from participants and each block contained few trials, GVS findings were treated as exploratory.

3 RESULTS

We first tested whether the optimization recovered a behaviour-linked signal with low consecutive-trial variability. We then characterized the anatomy of the resulting network, compared it with standard task activation, and examined medication- and GVS-related network effects.

3.1 Behavioural performance

After trial- and run-level quality control, 18 participants, 35 sessions, 70 runs, and 5,463 Go trials were retained for behavioural and neural analyses. Mean RT was 387.2 ± 65.8 ms in the OFF-medication state and 375.5 ± 64.7 ms in the ON-medication state. Consecutive-trial RT variability was 3.50 in the OFF state and 4.04 in the ON state. The subject-paired medication effect on mean RT was small and approached conventional significance in the supplementary analysis (mean ON–OFF difference = -13.6 ms, 95% CI $[-27.5, 0.3]$, $t(16) = -2.08$, $p = .054$; Supplementary Figure S2). Reward-condition effects were heterogeneous across subjects, and GVS produced no consistent group-level RT effect (Supplementary Figure S3). These behavioural results supported using trial-wise RT as the behavioural vigour index while treating medication and GVS behavioural effects cautiously.

3.2 The optimization identified a reproducible low-variability neural projection

The optimized voxel-weight maps were highly consistent across cross-validation folds (ICC = 0.978), indicating that the solution was not driven by a single fold. The resulting projection, y , was a one-dimensional weighted summary of voxel activity that jointly reflected task engagement, behavioural relevance, spatial coherence, and low consecutive-trial variability. We then asked whether this projected neural signal was less variable than the behaviour it was designed to track. Using subject-level summaries derived from the retained runs, behaviour-minus-projection variability was positive (mean difference = 7.5, 95% CI $[2.2, 12.8]$, paired $p = .008$; Figure 2A). Thus, the model recovered a neural signal that remained behaviour linked while changing more smoothly from trial to trial than RT itself.

We also tested whether the selected voxels had lower variability than a relevant motor-area comparison pool. The AAL3-derived motor control pool contained 44,176 voxels; after removing 1,823 overlapping selected voxels, 42,353 non-selected motor voxels remained before metric-validity filtering. The voxel-stability diagnostic used a metric-valid analysis mask and therefore was not expected to have exactly the same voxel count as the final p90 anatomical display mask. Within this diagnostic, the valid sets contained 9,050 selected voxels and 22,544 non-selected motor voxels, with no overlap. Selected voxels were enriched in the low-variability tail across percentile cutoffs (Figure 2B). Their mean consecutive-trial variability was 2.50, whereas 1,000 size-matched samples from the non-selected motor pool had a mean variability of 3.46 and a 95% CI of $[2.76, 4.38]$. The null sampling was matched by sample size only, not by hemisphere, ROI size, spatial location, or smoothness. This supports the conclusion that the optimized

network was enriched for stable voxels relative to a motor-area control pool rather than being an arbitrary sample of motor activity.

Together, these analyses show that the model recovered a reproducible, behaviour-linked neural projection with lower consecutive-trial variability than RT. This does not prove that all behavioural variability is downstream of the recovered signal, but it does show that a stable central vigour-related component can be detected in PD even when behaviour itself fluctuates.

3.3 The optimized vigour network was distributed across motor, associative, limbic, and subcortical regions

The final p90 vigour network contained 8,182 voxels and was distributed across both hemispheres (Figure 3). It was not limited to one primary motor cluster. The largest portions of the map were in temporal, parietal, and frontal cortex. A sensorimotor component was present in precentral, postcentral, paracentral, and supplementary motor areas. Additional clusters extended into orbitofrontal cortex, occipital cortex, fusiform and parahippocampal regions, hippocampus, amygdala, caudate, putamen, thalamus, and cerebellum.

ROI summaries confirmed that the map included motor and non-motor components. Temporal cortex contributed 25.27% of selected voxels, parietal cortex 13.46%, frontal cortex 11.38%, cerebellum 7.90%, and orbitofrontal cortex 7.22%. After normalizing by ROI size, stronger enrichment appeared in temporal-limbic and striatal regions, including parahippocampal gyrus, amygdala, caudate, orbitofrontal cortex, and temporal cortex. Motor-related areas, including supplementary motor area, paracentral lobule, precentral gyrus, parietal cortex, and cerebellum, were clearly represented but did not dominate the ROI-normalized profile (Supplementary Table S3 and Supplementary Figure S8). This anatomy is consistent with a vigour-related signal that integrates motor preparation with cue value, context, salience, and action organization.

3.4 The vigour network was not simply standard task activation

We compared the vigour network with the standard GLM task-activation map to test whether the optimized network was mainly a subset of general task activation (Figures 4 and 5). The two maps showed limited same-voxel overlap. The vigour network contained 8,182 voxels, whereas the task-activation map contained 21,610 voxels. Only 835 voxels were shared, giving a Dice coefficient of 0.056 and a Jaccard index of 0.029.

The task-activation map was broad and covered many regions involved in performing the task. In contrast, the vigour network was more selective. Some overlap appeared in motor-related regions, including pericentral cortex, supplementary/paracentral motor areas, putamen, cerebellum, and parietal cortex. However, several regions contained both maps with little or no same-voxel overlap, and temporal-limbic and orbitofrontal components were more prominent in the vigour network than in the task-activation reference (Figure 5; Supplementary Tables S4 and S5).

We also compared the vigour network with additional GLMsingle-derived activation maps. These analyses again showed low spatial overlap and distinct regional composition (Supplementary Figure S5). Together, these results show that the recovered network is not simply a smaller version of the task-activation map. The optimization captured a trial-wise vigour-related projection whose anatomy differs from the regions most consistently activated by the task on average.

3.5 Medication altered local network coupling

After defining the vigour network, we tested whether medication altered its task-based functional connectivity. The topology-control analysis showed that connectivity patterns differed more between medication states than between repeated runs within the same state, for both Pearson-based and mutual-information-based edge patterns (Supplementary Figure S9). This result indicates that medication state changed the organization of trial-wise coupling beyond ordinary run-to-run variability. It should not be interpreted as evidence that ON medication simply stabilized every selected voxel.

We next asked whether medication-related reorganization occurred locally within ROIs or between distinct ROIs (Figure 6). Medication primarily increased local connectivity within ROIs of the vigour network. Intra-ROI FC increased significantly from OFF to ON medication ($\Delta_{\text{ON-OFF}} = +0.0147$, $p = 0.006$),

whereas between-ROI FC showed no reliable change and remained close to zero ($\Delta_{\text{ON-OFF}} = -0.0006$, $p = 0.723$). This pattern suggests that dopaminergic medication strengthened local functional coupling within the vigour network rather than broadly increasing distributed connectivity between ROIs.

To assess whether this pattern was specific to the optimized network, we repeated the same analysis using the standard GLM task-activation map. The task-activation map also showed a significant difference between intra-ROI and between-ROI medication-related changes ($p = 0.012$; Figure 6). However, the individual effects were weaker. In the vigour network, the effect was driven by a robust increase in intra-ROI FC. In the task-activation map, intra-ROI FC showed only a non-significant increase ($\Delta_{\text{ON-OFF}} = +0.0081$, $p = 0.108$), together with a non-significant tendency toward reduced between-ROI FC ($\Delta_{\text{ON-OFF}} = -0.0098$, $p = 0.142$). Thus, task-active regions also captured a local-versus-distributed medication contrast, but the optimized vigour network more clearly identified medication-related strengthening of local coupling.

3.6 GVS produced exploratory connectivity changes within the recovered network

We next asked whether GVS changed functional connectivity within the final vigour-related network. For each active GVS condition, each edge was compared with the matched sham edge from the same subject, medication state, and run. Because each participant contributed multiple GVS blocks, observations were not fully independent. We therefore used this analysis to describe stimulation-sensitive edges rather than to make a strong therapeutic or group-level causal claim.

Using quantile mutual information, active GVS produced 156 FDR-significant sham-referenced edge changes out of 703 tested edges. Of these, 46 increased and 110 decreased (minimum $q_{\text{FDR}} = 0.0025$; Figure 7A). The strongest decreases involved Amygdala_L–Thalamus_L, Cerebellum_L–Putamen_L, and Cingulate_L–Thalamus_L. The strongest increases involved Parietal_L–Thalamus_L and Putamen_L–Supplementary Motor Area_L. At the anatomical-sector level, decreases were most concentrated between limbic/olfactory and somatomotor/cerebellar regions, with additional decreases linking subcortical, cingulate-temporal, and somatomotor/cerebellar sectors.

Using Spearman's rank correlation, active GVS produced 194 FDR-significant edge changes (Supplementary Figure S10). This pattern was more balanced, with 103 increases and 91 decreases (minimum $q_{\text{FDR}} = 0.0022$). The strongest increases involved Cingulate_R–Temporal_R, Fusiform_L–Putamen_L, and Fusiform_L–Precentral_R. The strongest decreases involved Amygdala_R–Precentral_L, Amygdala_R–Frontal_L, and Hippocampus_L–Temporal_L. The two metrics did not identify the same edge list, which is expected because they measure different forms of dependence. However, both analyses showed that GVS perturbed distributed interactions within the final network, repeatedly involving limbic/olfactory, medial-temporal, subcortical, visual, cingulate-temporal, and somatomotor/cerebellar components.

As a sanity check, we repeated the mutual-information analysis using the task-activation map instead of the optimized vigour network (Figure 7B). GVS also produced significant edge changes in this task-activation network. However, the detailed pattern was different: the task-activation map showed 134 significant mutual-information edges and 164 significant Spearman edges, with only 14 shared mutual-information edges and 16 shared Spearman edges overlapping the significant vigour-network edges. This supports a cautious interpretation: GVS affects task-active circuitry in general, but the optimized vigour-related network shows a partly distinct pattern of GVS-sensitive connectivity.

4 DISCUSSION

This study asked whether trial-wise variability in motor vigour in PD reflects instability in a central vigour-related neural signal, or whether some behavioural variability arises downstream of a relatively stable central signal. The results support the second interpretation. Using fMRI during a ballistic hand-squeeze task, we identified a distributed neural projection that tracked RT while showing lower consecutive-trial variability than the behavioural signal itself. The selected voxels were also less variable than a broader motor-area comparison pool. The main finding is therefore not simply that RT was associated with brain activity. Rather, the study recovered a behaviour-linked neural expression of vigour in PD that was more stable than the behaviour it tracked.

This dissociation is important because behavioural variability alone cannot reveal where instability enters the motor system. In PD, response times fluctuate from moment to moment, and motor-unit firing can be more variable even when average performance is partly preserved (MacDonald et al., 2025; Wilson et al., 2020). More generally, movement variability can arise from central preparatory states, adaptive exploration, and downstream motor noise (Churchland et al., 2006; Faisal et al., 2008; Harris and Wolpert, 1998; Pekny et al., 2015; Wu et al., 2014). The present findings argue against a simple interpretation in which variable vigour directly reflects an unstable central vigour command. Instead, they suggest that a relatively stable central signal related to trial-wise vigour can still be recovered in PD, and that at least some behavioural fluctuation in movement initiation may arise downstream of this signal or through its interaction with later motor processes.

A potential concern is that this conclusion could appear circular because low consecutive-trial variability was one optimization criterion. However, the analysis did not merely impose stability and then report the same property as a result. The model was constrained not only by stability, but also by task engagement, behavioural relevance, and spatial coherence. We therefore tested the recovered solution against meaningful baselines. At the subject level, the optimized projection was less variable than RT. At the voxel level, the selected voxels were concentrated among low-variability voxels and showed lower mean variability than size-matched samples from non-selected motor-area voxels. These validation analyses show that the solution was not an arbitrary smooth sample of motor activity and was not simply a restatement of behavioural variability.

The spatial organization of the projection further supports this interpretation. The recovered vigour-related network included a motor-related core with contributions from pericentral cortex, superior parietal cortex, putamen, and cerebellar regions. This is consistent with prior work implicating motor cortical, basal-ganglia, thalamic, and cerebellar systems in movement speed, force, amplitude, and vigour (Anderson et al., 2012; Dudman and Krakauer, 2016; Shirinbayan et al., 2019; Spraker et al., 2007; Turner and Desmurget, 2010; Yttri and Dudman, 2016). However, the projection was not limited to execution-related motor regions. It also included amygdala, hippocampus, parahippocampal cortex, temporal cortex, frontal and orbitofrontal cortex, and striatum. A signal reflecting pure motor execution would be expected to localize more strongly to primary motor and adjacent execution-related areas. In contrast, the present projection was enriched in regions positioned to integrate cue value, context, salience, motivational state, and action readiness.

This network organization fits the behavioural demands of the task. The putamen and broader basal ganglia are directly relevant to vigour because dopaminergic cortico-basal-ganglia circuits regulate movement velocity and energetic investment in action (Dudman and Krakauer, 2016; Niv et al., 2007; Turner and Desmurget, 2010; Yttri and Dudman, 2016; Zénon et al., 2016). Amygdala involvement is also plausible because primate amygdala neurons encode positive and negative visual-stimulus value during learning (Paton et al., 2006), and anatomical evidence supports an amygdala-motor pathway through which affective or salient cues can shape action (Grèzes et al., 2014). Hippocampal and parahippocampal involvement may reflect contextual organization of action: reward behaviour depends on hippocampus-nucleus accumbens synaptic strength (LeGates et al., 2018), the hippocampus can organize upcoming skilled action sequences (Yewbrey and Kornysheva, 2024), and hippocampal mechanisms support visually cued contextual response selection (Kim et al., 2012). Together, these findings suggest that the recovered network integrates motor preparation with the contextual and motivational signals that determine whether an action is initiated quickly on a given trial.

The limited overlap with the task-activation map strengthens this argument. A task-activation map asks where activity is reliably evoked by performing the task. The optimization asked a different question: which spatially coherent neural projection is linked to how quickly an action is initiated while maintaining a stable expression over consecutive trials? The low Dice and Jaccard values indicate that the network most consistently activated by the task is not the same as the network that best captures stable trial-wise vigour-related coding. This distinction matters because it separates mean task activation from trial-wise vigour coding. Vigour is therefore not simply a property of task-evoked activation amplitude; it is expressed through the organization of a distributed neural signal linking preparatory, motivational, contextual, and motor systems.

The medication findings add a state-dependent component to the same story. Dopaminergic medication changed trial-wise network organization and primarily strengthened local within-ROI coupling in the

vigour network. This effect was stronger and clearer in the optimized network than in the standard task-activation reference. The result should not be overstated as proof that dopamine normalizes the network or increases cross-subject convergence, because the medication session order was not counterbalanced and no healthy-control group was included. A safer interpretation is that dopaminergic state alters the internal organization of the recovered vigour-related network, especially the local coherence of voxel-level responses within ROIs. This is consistent with prior work linking dopamine to movement vigour and with evidence that levodopa can alter functional network organization in PD (Niv et al., 2007; Berman et al., 2016; Zénon et al., 2016; Mazzoni et al., 2007).

The medication result also shows why a purely downstream-noise account is incomplete. If behavioural variability arose only from noise added after a stable central command, dopaminergic medication might improve behavioural output without changing the central network that tracks trial-wise vigour. Instead, the ON-state result suggests that dopamine also acts on the organization of the behaviour-linked central signal itself. Thus, some behavioural variability may arise downstream of central vigour-related coding, but dopamine appears to shape the network through which that coding is expressed.

The GVS findings provide exploratory perturbation evidence. Active stimulation produced sham-referenced connectivity changes within the final network, but the pattern was heterogeneous across stimulation waveforms, medication states, subjects, and edge metrics. The value of this result is not that GVS restored a common vigour-network configuration. It did not. Instead, GVS showed that components of the recovered network are perturbable by vestibular stimulation, and that this perturbation is state- and metric-dependent. This interpretation is consistent with previous work showing that GVS effects in PD depend on stimulation frequency, individual physiology, and parameter selection (Lee et al., 2021; Cai et al., 2018; Menon et al., 2025). The practical implication is that future stimulation studies should move beyond testing whether stimulation changes mean activation in selected ROIs. A more informative approach would ask whether individualized stimulation parameters alter the stability, topology, or behavioural coupling of the vigour-related network.

Several limitations define the scope of this interpretation. First, the study did not include a healthy control group. The results therefore cannot determine whether the recovered network is specific to PD or whether it reflects a more general vigour-related signal that remains detectable in PD. Second, RT was used as the primary trial-wise index of vigour because it was the most robust behavioural measure in this dataset. RT is highly relevant to response initiation, but vigour also includes movement speed, force amplitude, force rise time, and movement duration. Future studies should test whether the same network tracks multiple components of vigour or whether distinct but overlapping networks support initiation speed and force production.

Several design features limit causal interpretation. The medication sessions followed a fixed OFF-then-ON order; medication-related differences may therefore partly reflect practice, fatigue, time-on-task, scanner drift, or other session-order effects. We interpreted medication findings as state-associated rather than fully randomized medication effects, but a counterbalanced medication order would provide stronger evidence. The first three participants also had a different functional MRI acquisition protocol from the rest of the sample. This may have affected signal-to-noise, temporal sampling, or single-trial beta estimation. Future replication should use a single acquisition protocol across all participants.

The task structure also means that RT-linked neural signals cannot be attributed only to motor vigour. Reward cues, GVS condition, block context, and adaptation across the run may all influence both RT and trial-wise neural activity. These factors were part of the experimental context and were handled cautiously in the analysis, but they could still contribute to the recovered projection. The optimization framework was also linear. This improves interpretability and supports identification of a consistent low-variability axis, but nonlinear relationships between distributed brain activity and vigour may be missed. Finally, the same dataset was used both to identify the network and to characterize its properties. Cross-validation and control analyses reduce, but do not remove, the need for external validation in an independent PD cohort, healthy adults performing the same task, and PD participants performing a different motor task.

Overall, the present study suggests that trial-wise vigour variability in PD should not be interpreted simply as instability of a central motor command. A relatively stable, behaviour-linked neural signal can be recovered, and this signal is organized across motor, striatal, limbic, contextual, and motivational systems. Behavioural variability may therefore partly arise downstream of central vigour-related coding,

but dopaminergic state and vestibular stimulation also modulate the network through which this coding is expressed.

5 CONCLUSIONS

The present study identifies a distributed vigour-related neural projection in Parkinson's disease whose trial-wise expression is more stable than the behavioural reaction-time fluctuations it tracks. This projection is anatomically distributed, differs from standard task activation, and shows dopaminergic-state-related strengthening of local within-ROI coupling. Exploratory GVS analyses further show that the recovered network is perturbable by vestibular stimulation, although these effects are heterogeneous and should not be interpreted as evidence for a single therapeutic stimulation pattern. Together, the findings support a model in which PD-related variability in response initiation is not solely the product of an unstable central signal. Instead, a relatively stable central vigour-related signal can be recovered, while behavioural variability may arise partly as that signal is transformed into motor output under different dopaminergic and stimulation states.

ETHICS STATEMENT

The studies involving human participants were reviewed and approved by the Clinical Research Ethics Board at the University of British Columbia. The participants provided written informed consent to participate in this study.

CONFLICT OF INTEREST STATEMENT

[Please confirm: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.]

AUTHOR CONTRIBUTIONS

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SUPPLEMENTAL DATA

Supplementary material for this article is provided in the accompanying Supplementary Material file.

DATA AVAILABILITY STATEMENT

[Please complete: The datasets generated and/or analyzed during the current study are available in repository name/link, or state the appropriate restriction if the data cannot be publicly shared.]

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FIGURE CAPTIONS

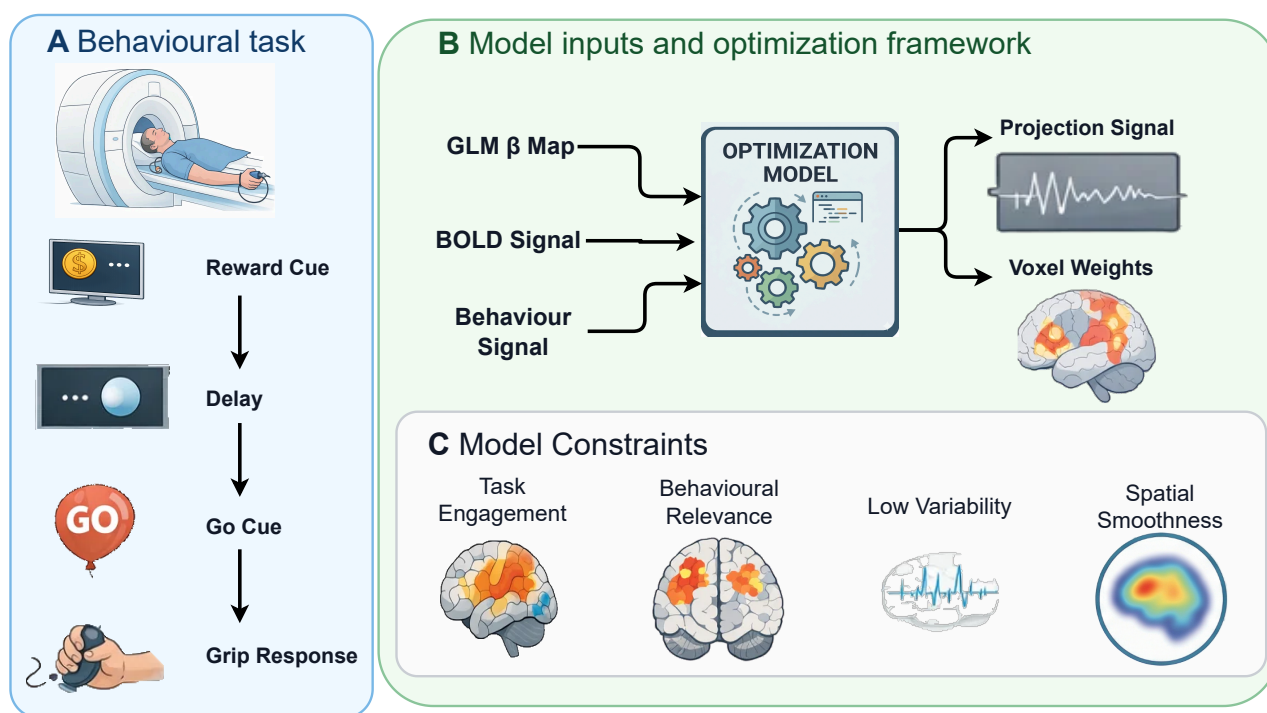


Figure 1. Overview of the behavioural task and analysis pipeline. **A**, Participants performed a reward-cued ballistic hand-squeeze task during fMRI. Each trial progressed from reward cue to variable delay and then to a Go signal that prompted a rapid grip response; reaction time and whole-brain BOLD signals were recorded during the task. **B**, Trial-wise neural responses were estimated with GLMsingle to obtain single-trial beta maps and BOLD-derived features. These neural features and reaction-time measures entered the constrained optimization framework, yielding a low-dimensional projection signal and a voxel-weight map. **C**, The optimization searched for a neural projection that was task engaged, behaviourally relevant, spatially coherent, and relatively stable across consecutive trials.

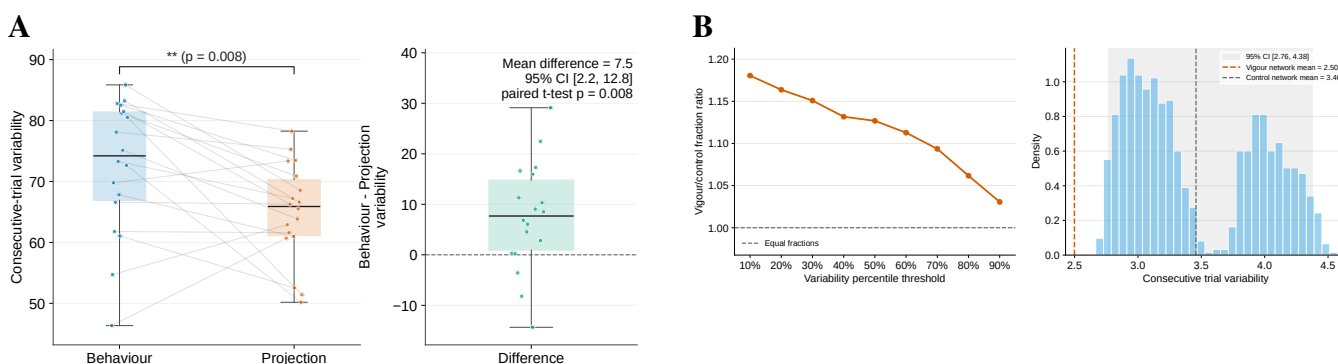


Figure 2. Validation of the optimization framework. **A**, Consecutive-trial variability was lower for the optimized neural projection than for reaction time. Positive behaviour-minus-projection differences indicate that the projection changed less abruptly than behaviour; paired inference used subject-level summaries derived from the retained runs. **B**, Selected voxels showed lower consecutive-trial beta variability than size-matched samples from the non-selected motor-area control pool and were enriched among low-variability voxels.

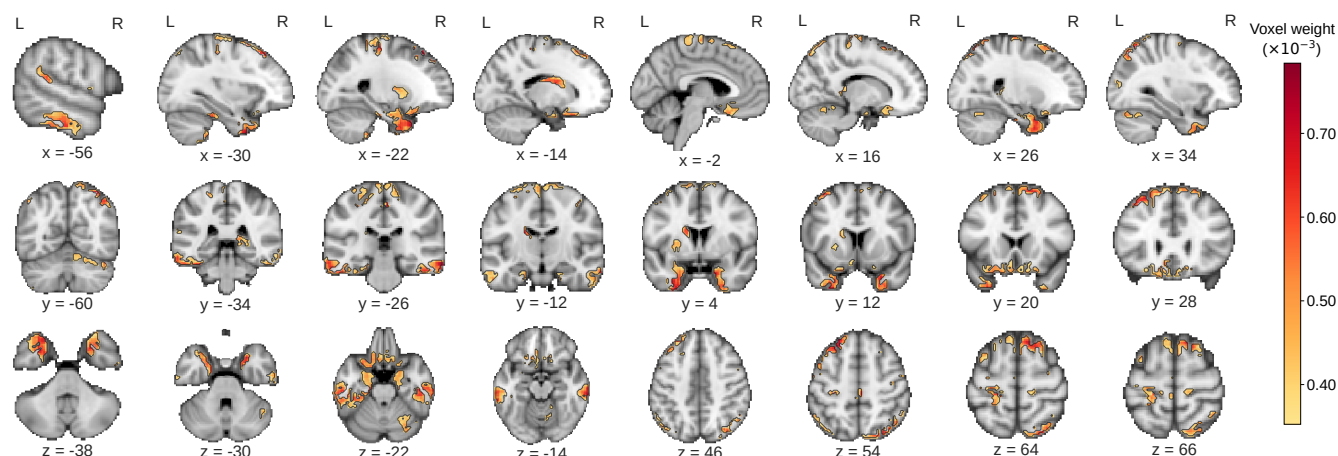


Figure 3. Spatial distribution of the optimized vigour-related network. Thresholded voxel weights are overlaid on representative sagittal, coronal, and axial MNI slices. Warmer colors indicate larger model-derived voxel contributions, scaled by 10^{-3} .

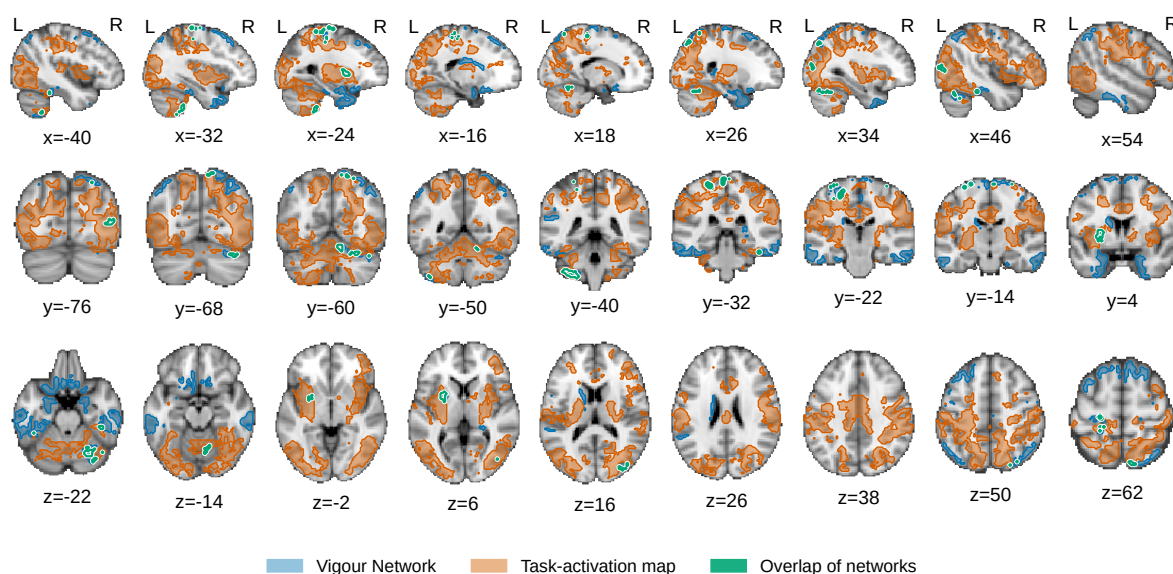


Figure 4. Voxelwise overlap between the optimized vigour network and the standard task-activation map. Blue shows the vigour network, orange shows the task-activation map, and green marks same-voxel overlap across representative brain slices.

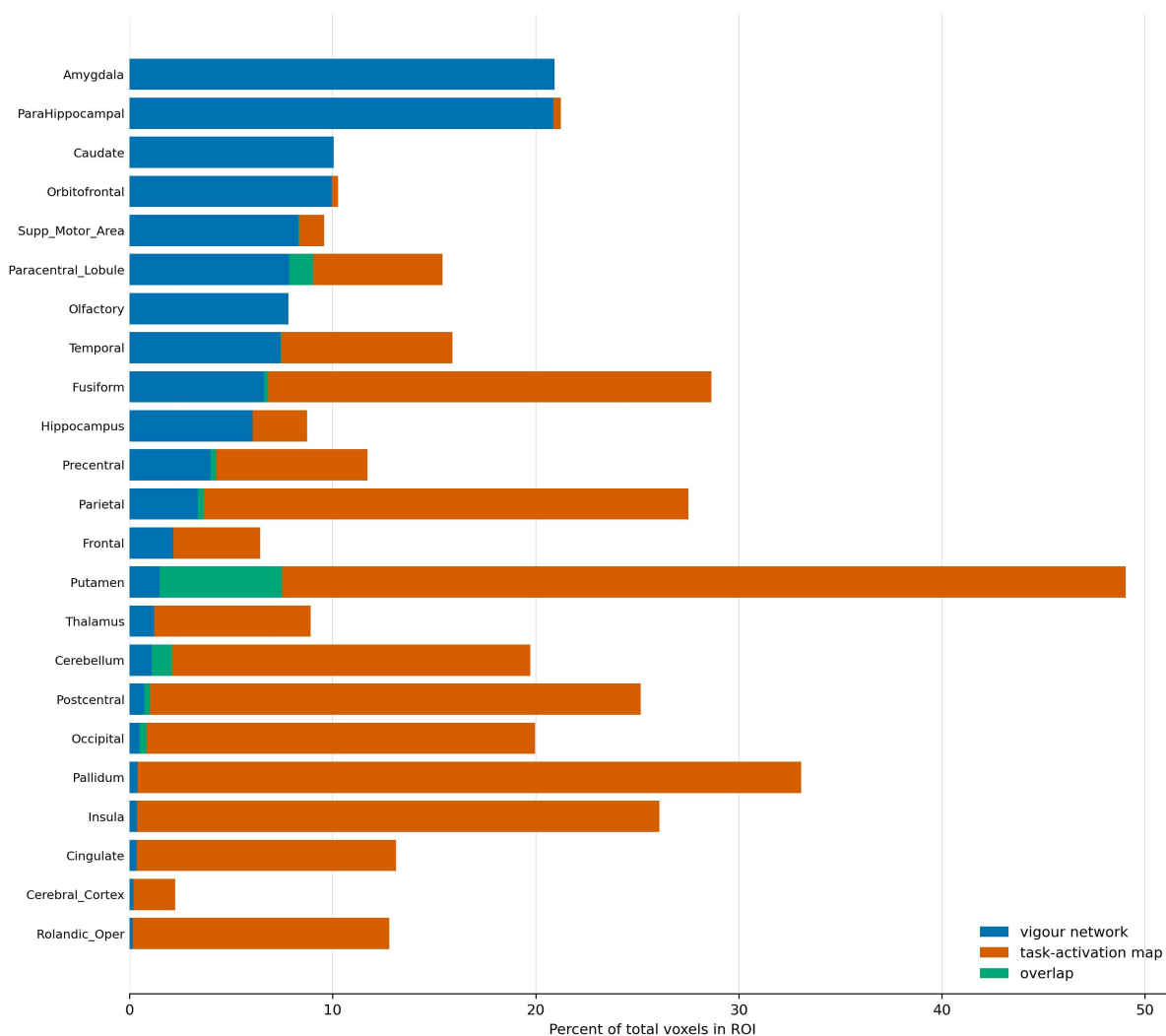


Figure 5. ROI-wise anatomical comparison of the optimized vigour network, the standard task-activation map, and their same-voxel overlap. Horizontal bars show the percentage of each ROI occupied by the vigour network, the task-activation map, and overlapping voxels. The vigour network is relatively enriched in temporal-limbic, orbitofrontal, and striatal regions, whereas the task-activation map occupies broader sensorimotor, cerebellar, parietal, and subcortical regions.

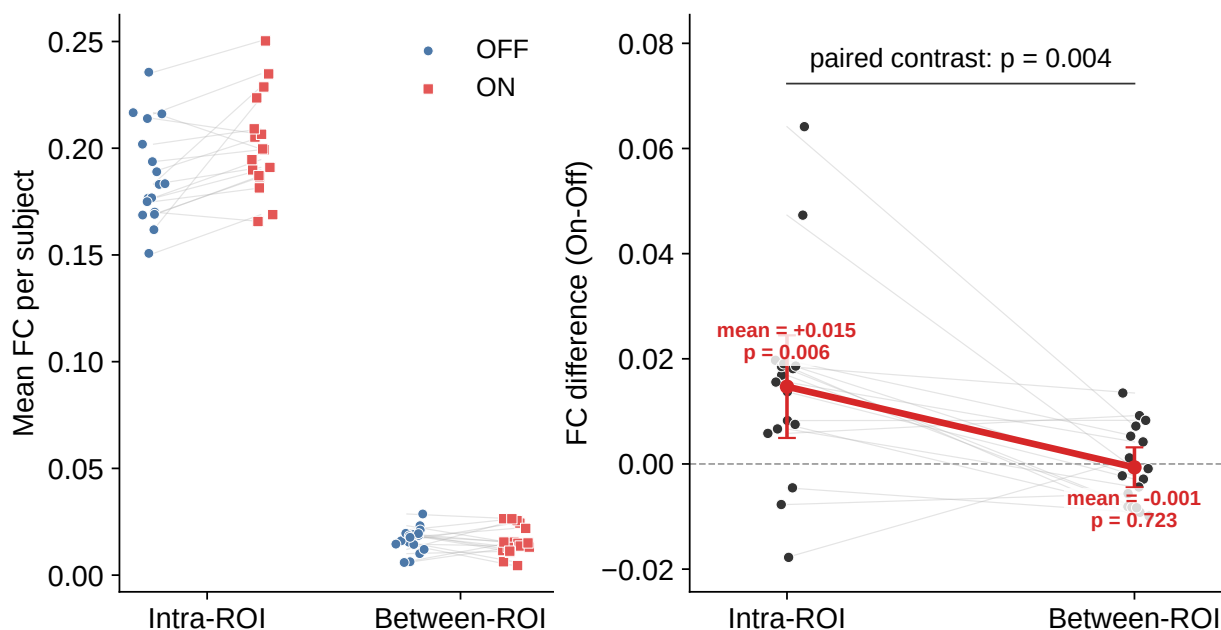
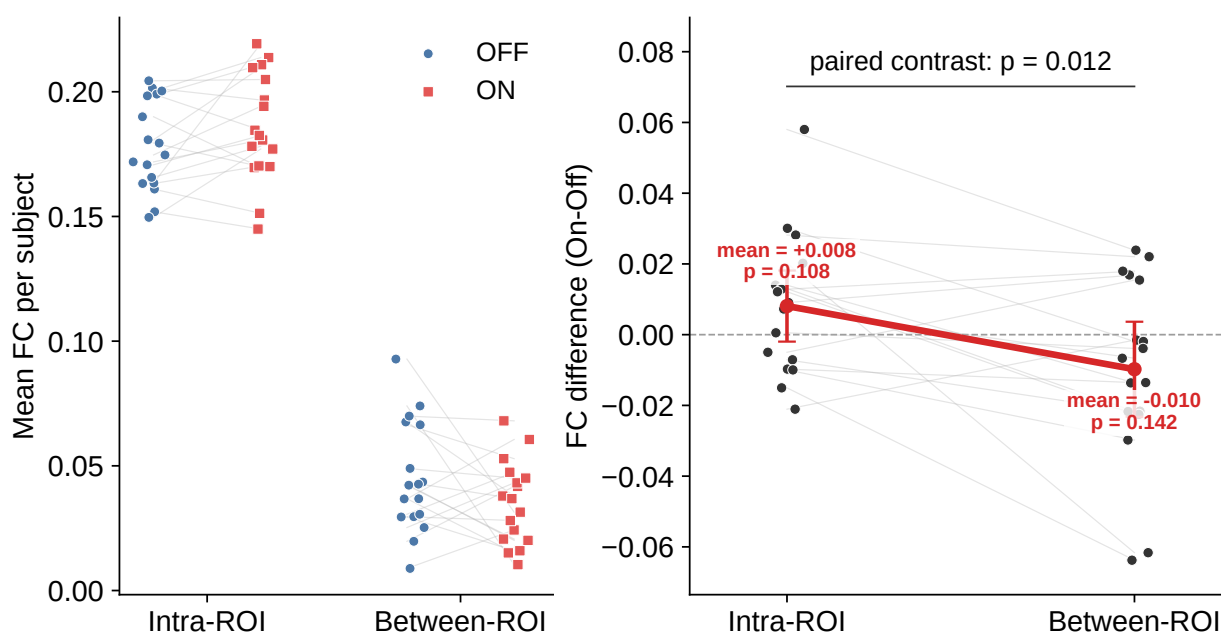
A**B**

Figure 6. Medication effects on local and distributed functional connectivity. **A**, Within-ROI and between-ROI Pearson functional connectivity for the optimized vigour network, shown for OFF and ON medication and as participant-level ON–OFF differences. Medication significantly increased intra-ROI coupling, whereas between-ROI coupling remained close to zero. **B**, The same analysis applied to the standard task-activation map. The intra-ROI versus between-ROI contrast was significant, but the individual intra-ROI and between-ROI changes were weaker than in the optimized vigour network.

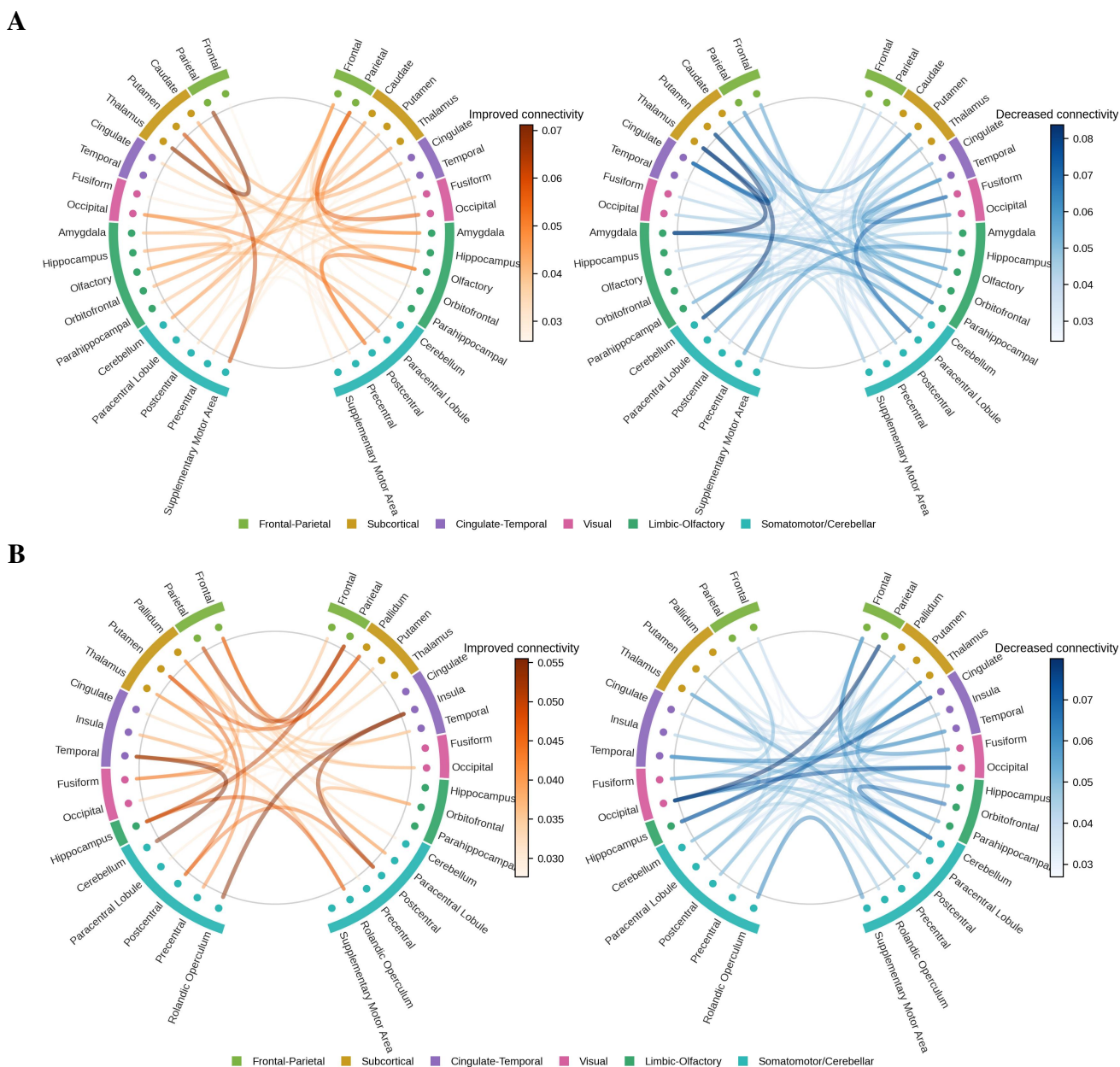


Figure 7. Exploratory GVS-related mutual-information edge changes in the optimized vigour and task-activation networks. **A**, FDR-significant active-minus-sham mutual-information edge changes within the optimized vigour network, pooled across subjects, medication states, runs, and active GVS waveforms. Orange edges indicate increased statistical dependence and blue edges indicate decreased dependence; color intensity scales with mean edge-change magnitude. **B**, The same mutual-information analysis in the task-activation network, showing a partly distinct pattern of GVS-sensitive edges. Nodes are labeled by ROI and grouped by anatomical sector.